A Concise Title for Your Preprint: Optional Subtitle Describing the Contribution

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This is the CAB Lab preprint template. It provides a boxed title panel in translucent Princeton orange with the lab logo, author and affiliation lists, an abstract, keywords and resource links, followed by a single-column body with sans-serif headings, styled captions and a running header. Replace this paragraph with a summary of the problem, the key idea, the evaluation setting and the main result. Keep the abstract to a single paragraph of roughly 150–250 words so that the title panel fits comfortably on the first page together with the opening of the introduction.

Keywords geometric deep learning, equivariance, AI for science, robotics

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Code https://github.com/CAB-Lab-Princeton/your-project

Project https://cab-lab-princeton.github.io/your-project

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1 Introduction

Start with the problem and why it matters. Symmetry is one of the most reliable inductive biases in machine learning for the physical sciences: a model whose outputs transform predictably under rotations, translations or permutations needs less data and generalizes further [1, 2]. Figure 1 shows how a figure, its caption and a cross-reference look in this template.

Contributions. State them as a short list.

- A new model, GeoNet, that is equivariant by construction (Section 2).
- A theoretical guarantee on its expressivity (Theorem 1).
- Experiments on simulated and real robotic systems (Section 3).

1.1 Related work

Group related work by theme in short paragraphs, cite with `\cite` [3], and close with one sentence on how the present work differs.



Figure 1 | Teaser figure. Replace this placeholder with the figure that best summarizes the paper. Captions open with a bold one-line takeaway, then explain what the reader should look at.

Algorithm 1 Training loop of GeoNet

Require: dataset $\mathcal{D}$, learning rate η
1: **for** each minibatch $(x, y) \in \mathcal{D}$ **do**
2: $\quad \hat{y} \leftarrow f_\theta(x)$
3: $\quad \theta \leftarrow \theta - \eta \, \nabla_\theta \ell(\hat{y}, y)$
4: **end for**

2 Method

Keep paper-specific notation in `main.tex`. Displayed equations are numbered and can be referenced with `\cref`, as in Equation (1):

$$f(\rho_{\text{in}}(g)\, x) = \rho_{\text{out}}(g)\, f(x), \qquad \forall g \in G. \tag{1}$$

Theorem 1 *Every layer of GeoNet commutes with the action of* E(3)*, hence so does the composed network.*

Implementation. Run-in paragraph headings use `\paragraph` or `\parahead`. Algorithm 1 shows the algorithm environment.

3 Experiments

Tables use `booktabs`; the helper `\tablestyle{<colsep>}{<stretch>}` sets compact spacing. Captions of tables sit above the table (Table 1).

Discuss what the numbers show, then add ablations and failure cases.

4 Conclusion

Summarize the contribution, state the main limitation honestly, and point to future work.

Table 1 | Main results. Mean ± std over five seeds; best in bold.

Method	Equivariant	Params	Error (↓)	Time (s)
MLP	✗	1.2M	0.412 ± 0.010	0.8
Transformer	✗	3.4M	0.231 ± 0.008	2.1
EGNN	✓	0.9M	0.118 ± 0.004	1.3
GeoNet (ours)	✓	0.8M	**0.087 ± 0.003**	1.1

Acknowledgments

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References

[1] Michael M. Bronstein, Joan Bruna, Taco Cohen, and Petar Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv preprint arXiv:2104.13478*, 2021.

[2] Taco Cohen and Max Welling. Group equivariant convolutional networks. In *International Conference on Machine Learning*, pages 2990–2999, 2016.

[3] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.

A Additional details

Appendix sections are lettered automatically after `\beginappendix`.